The Effect of Industry Experience on Hypothesis Generation and Audit Planning Decisions

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Abstract

A significant concern in behavioral research in accounting has been the effect of experience on judgment and decision-making. As widely recognized, there are, however, several dimensions to experience, including general domain experience and task-specific experience. One important dimension that has received limited attention is industry experience. For instance, in an audit context, greater industry experience is expected to lead to greater effectiveness and efficiency as auditors develop a knowledge-base of the unique risks and audit approaches for a particular industry. The purpose of this study is to investigate the impact of industry experience on the generation of hypotheses of likely errors in conducting analytical procedures. Other audit planning tasks are also examined (e.g., risk assessment and extent of testing).

Seventy-two auditors, 34 with significant retailing experience and 38 without such experience (both groups ranging in rank from senior to partner), completed a comprehensive, realistic case for a retailing client. Four material errors, three relating to a retailing environment, were present in the case. The findings indicate that industry experience significantly enhanced hypotheses generation in identifying errors but did not result in expected risk assessments or revisions to planned extent. Proportionately greater audit hours were, however, assigned to more experienced audit staff for misstated accounts.

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THE EFFECT OF INDUSTRY EXPERIENCE ON HYPOTHESIS GENERATION AND AUDIT PLANNING DECISIONS

ABSTRACT

A significant concern in behavioral research in accounting has been the effect of experience on judgment and decision-making. As widely recognized, there are, however, several dimensions to experience, including general domain experience and task-specific experience. One important dimension that has received limited attention is industry experience. For instance, in an audit context, greater industry experience is expected to lead to greater effectiveness and efficiency as auditors develop a knowledge-base of the unique risks and audit approaches for a particular industry. The purpose of this study is to investigate the impact of industry experience on the generation of hypotheses of likely errors in conducting analytical procedures. Other audit planning tasks are also examined (e.g., risk assessment and extent of testing).

Seventy-two auditors, 34 with significant retailing experience and 38 without such experience (both groups ranging in rank from senior to partner), completed a comprehensive, realistic case for a retailing client. Four material errors, three relating to a retailing environment, were present in the case. The findings indicate that industry experience significantly enhanced hypotheses generation in identifying errors but did not result in expected risk assessments or revisions to planned extent. Proportionately greater audit hours were, however, assigned to more experienced audit staff for misstated accounts.

In recent years auditing firms have increasingly specialized staff into industry areas (Danos and Eichenseher, 1986; Craswell and Taylor, 1991; Emerson 1993). It is reasonable to expect that industry experience should enhance audit effectiveness and efficiency. Error patterns and, hence, areas of exposure vary by industry (Wright and Aston, 1989; Kreutzfeldt and Wallace, 1986; Maletta and Wright, 1996). Further, as evidence of greater effectiveness, fewer violations of governmental auditing standards have been found for CPA firms with greater specialization in that industry (O'Keefe et. al., 1994). Lastly, industry specialization may be utilized to establish differentiation in service quality and as a marketing tool to attract clients (e.g., KPMG Peat Marwick; Emerson 1993). Findings by Craswell et. al. (1995) suggest these efforts are successful in creating recognition in the audit market of the value of industry expertise. They report that clients on average paid a 14% premium for industry specialization over and above the premium for Big 8 name recognition. Elder (1994) also finds lower levels of underpricing and higher cash compensation when industry specialist auditing firms are

involved in an initial public offering. Despite trends toward greater specialization, we have limited knowledge of the effect of industry experience on audit judgments.

The primary purpose of this study is to examine the impact of industry experience on hypothesis generation in the audit planning phase. Identification of likely errors at this stage has an important impact on audit effectiveness and efficiency (Libby, 1985; Houghton and Fogarty, 1991).

The remainder of the paper is divided into four sections. The next section contains a review of the relevant literature and an identification of the research hypotheses. The two sections to follow provide an overview of the method employed and the results. The final section is devoted to a discussion of the results and their implications for practice and future research.

RELEVANT LITERATURE AND RESEARCH HYPOTHESES

Prior research has distinguished between general audit ("domain") experience, task experience, and domain-specific experience. General audit experience relates to broad knowledge of auditing through longevity in the field. Conflicting findings regarding the effects of general audit experience led Abdolmohammadi and Wright (1987) and Bonner (1990) to examine the role of task experience, since task experience may not match precisely with general experience. For example, staff accountants often receive significant task experience in evaluating internal controls. Thus, increased general experience beyond this level may not improve the performance of an individual in performing this task. The findings of these two studies indicate that task experience significantly impacts auditor judgments and decisions. Further, Choo and Trotman (1991) find that auditors with greater task experience exhibit an enhanced knowledge structure and ability to make inferences.

Domain-specific experience relates to familiarity with an area of auditing that requires specialized knowledge such as particular accounts or industries. For example, Bedard and Biggs (1991) report that auditors with greater experience with clients having inventory and in the manufacturing area, the setting in which their experimental case was based, were more likely to identify the error present in a complex analytical procedure task. Greater domain-

specific experience is also found to enhance the planning of appropriate audit tests (Bedard, Biggs, and DiPietro, 1993; Bedard and Wright, 1994).

Recently the importance of industry experience has been recognized as a separately identifiable dimension of domain-specific experience (Solomon et. al., 1996; Libby, 1993). Despite its emphasis in practice, there has been limited empirical research that considers the role of industry experience in audit judgments. In a concurrent partner review task involving six subjects, Johnson et. al. (1991) report that industry experience is of greater value in identifying the presence of a material fraud than general audit experience. Bonner and Lewis (1990) indicate superior analytical procedure performance for auditors with specialized industry experience. However, Ashton (1991) did not find significant industry effects in accuracy of error frequency knowledge.

Taylor (1994) investigates auditor inherent risk judgments of banking and non-banking specialists. A banking case was utilized where inherent risks where varied (high or low) in two account areas: one unique to banking (loans receivable) and another that is generic (premises and equipment). Second-order uncertainty (confidence and variance in confidence) was also examined. The results provide limited support that industry specialization enhanced auditor sensitivity to risk variations. However, the strongest experimental effect was a pronounced higher level of second-order uncertainty and resulting conservatism, as hypothesized, for the non-banking group.

Finally, Solomon et. al. (1996) examine the effect of industry specialization on error frequency knowledge and the subsequent planning of audit tests. Experienced auditors (managers and partners) specializing in either the health-care or the financial services industries were provided with a low likelihood, industry-specific non-error explanation for material fluctuations encountered in a set of ratios of a hypothetical client. Using a within subjects design, participants received both a case in their area of industry specialization and another case in an unfamiliar industry. The results indicated that industry experience led to greater accuracy in initially assessing the likelihood of the non-error explanation when compared to a panel of industry experts, suggesting the importance of knowledge of business operations. Industry specialization also led to enhanced content and completeness of knowledge as indicated by asking for a revised assessment of the likelihood of the explanation

given two additional explanations. Auditors specializing in the industry revised their initial belief significantly less than those without substantial industry experience. This finding was anticipated to occur as a result of a more complete set of hypotheses contained in the knowledge base of specialists than non-specialists. Thus, the additional explanations had little information content to the specialists. A second experiment revealed industry experience resulted in less propensity to follow a confirmatory testing strategy when provided with a management explanation.

The Role of Industry Experience in Hypothesis Generation

Across a wide variety of diagnostic settings, prior studies have shown that hypothesis generation significantly impacts subsequent evidence search and decision accuracy (Gettys et. al., 1986; Mehle, 1982; Einhorn and Hogarth, 1981). For example, Elstein et. al. (1978) find that physicians who fail to initially consider the correct illness are subsequently unable to determine the appropriate diagnosis. Further, Gettys and Fisher (1979) report that improper hypothesis generation can result in lack of consideration of alternative causes, resulting in decision errors. In an analytical procedures context, greater general audit experience has been associated with the generation of a larger number of total hypotheses and plausible hypotheses (Libby, 1985; Libby and Fredrick 1990).

Building on the work of Johnson et. al. (1991), industry-specific experience is expected to improve hypothesis generation in accurately identifying industry related errors that may be present. A recent experimental study by Whittington et. al. (1995) provides evidence that industry experience enhances auditors' abilities to generate plausible hypotheses for a client in their area of specialization but not for other industry settings. Tuttle (1996) also reports that when specific patterns of unexpected fluctuations are encountered auditors use their knowledge gained from experience to recall and evaluate the frequency of plausible errors. Industry experience is expected to aid auditors in considering unique error patterns associated with an industry. Finally, from an extensive review of the literature, Koonce (1993) observes that in the analytical review task the auditor often possesses substantial specialized knowledge and experience. However, industry experience would not be anticipated to be beneficial in

identifying generic errors that are not unique to the industry setting of the client. Accordingly, H1 & H2 investigate the effect of industry experience on hypothesis generation.

H1 Auditors with greater industry-specific experience generate a greater number of plausible hypotheses.

H2A Auditors with greater industry-specific experience generate a greater number of hypotheses identifying the presence of unique industry errors.

H2B Auditors with greater industry-specific experience do not demonstrate superior hypothesis generation performance in identifying a generic error.

Risk Assessment and Audit Extent Judgments

In addition to hypothesis generation, two other audit tasks that may be effected by industry experience are examined: inherent risk assessments; and the planning of audit extent. Prior archival research (Maletta and Wright, 1996) demonstrates unique pattern of errors present for various industries, suggesting that industry experience should enhance recognition of risk areas.

H3: Industry experience improves an auditor's ability to identify accounts containing material errors.

Industry experience is anticipated to lead to greater effectiveness in program planning. If, as expected, industry experience leads to improved identification of likely errors (H1 and H2) and risk assessment (H3), the program plan may then be appropriately tailored to areas (e.g., accounts) of risk exposure. That is, testing may be increased where the likelihood of risk appears high. These expectations lead to H4.

H4: The planned level of audit testing at the account level is more closely related to the presence of material errors as industry experience increases.

Audit costs are not only dictated by the number of hours planned but also how these hours are assigned across staff levels. For instance, senior or supervisor time is more costly than staff time. Greater industry experience should aid the auditor in assigning staff according

to risk exposure. To control audit risk, more experienced staff are expected to be directed to areas of significant risk of a material error to ensure audit effectiveness. Consequently, H5 posits that industry experience will impact the assignment of staff.

H5: Industry experience is positively associated with hours assigned to more experienced staff for misstated accounts.

METHOD

Overview

An experiment was conducted using practicing auditors and a planning task from an audit of a disguised retailing client. Four material errors were embedded in the case, three specific to retailing. Participating auditors were asked to perform three planning tasks: assessing the likelihood of a material error by account; determining audit extent (by staff level); and generating hypotheses. Since practicing auditors do not formally identify hypotheses during the planning stage (Hirst and Koonce, 1996), directing subjects to identify errors earlier in the experiment may have artificially altered planning judgments. Consequently, the hypothesis generation task was placed at the end of the experiment to avoid potentially biasing the risk assessment and audit extent judgments.

Subjects

Subjects for the experiment were seventy-two auditors randomly selected from two pools, those with significant retailing experience and those without such experience. The research instrument was distributed by a senior audit partner in each of two participating Big 6 firms with a cover letter of support[1]. Subjects were directed to perform the tasks individually and were assured confidentiality of responses. As reflected in Table 1, both groups had approximately the same level of general audit experience (about 8 years), distributed across the levels of supervisory/senior, manager, and partner. As expected, retail specialists had a significantly ($p < .001$) greater number of retailing engagements, clients, and training[2]. Since the study examines general planning tasks that auditors have some exposure to after 2-3 years of fieldwork, subjects are at the senior or high levels, representing individuals who have the necessary task experience to make the planning judgments studied.

The Experimental Case

As noted, the industry setting examined is retailing. This industry was selected based on discussions with senior audit partners of the participating firms. The retailing industry is unique enough to require industry knowledge yet not so specialized (e.g., banking) that engagement planning could not be reasonably considered by most auditors. Furthermore, many firms specialize in retailing (Craswell and Taylor, 1991), as do both firms who participated in the study.

The case materials contained seven pages of background information and two pages of summary, comparable financial data which all subjects received.[3] Background information revealed that the client was a publicly-held, national retailer who had recently acquired a chain of 20 stores which brought the company's total to 110 stores. Although individual stores retained substantial control over buying and pricing decisions, the client relied heavily upon centralized purchasing and warehousing functions. The client was currently developing an automated accounts payable and expense payables matching system. The client had also recently initiated inventory cycle counts whereby counts were conducted for specified departments throughout the year rather than at year end. The case materials were developed with extensive involvement by a retailing partner and careful pretesting by 10 auditors.[4]

Four material errors were embedded in the case:

markdown omissions
inventory shrinkage
lower-of cost-or-market
improper expense payables cut-off.[5]

The national director of the retailing practice of one of the firms was asked to identify a retailing engagement where there were material errors detected that were unique to the industry. The first three errors were considered to be of this type.

Clearly, markdowns are only present for retailing clients. Although inventory shrinkage[6] and lower-of-cost-or-market write-downs are not unique to retailing, the circumstances that led to these errors are. First, because of the Retail Method of accounting, the level of shrinkage for a retailer entails additional, unique estimation problems from that of other clients. Secondly,

for the experimental case the shrinkage error (to be described more fully) occurred as a result of a change in the manner in which the physical inventory counts were taken from a count at year end to interim cycle counts. Thus, identifying this error required knowledge of the Retail Method and how shrinkage is estimated as well as the likely problems associated with cycle counts in a retail environment (e.g., frequent variations in shrinkage rates). In addition, the realism of the retailing errors was enhanced by drawing them directly from the actual disguised audit and through discussions with retailing partners.

The fourth error is generic in that it may exist in many industry environments. The extensive client background information, financial data, and ratios contained in the case provided cues suggesting the presence of all four errors.

The first two retailing errors relate to overstatements of inventory .The first error resulted from initiation of markdowns at the store level. Although ratio data revealed no change in markdowns, the background information suggested that markdowns should have been higher due to declining market conditions. Markdown errors occurred due to a failure by store managers to notify accounting at the corporate level of markdowns initiated, a common problem of a decentralized markdown system. Further, inventory shrinkage was underestimated. Shrinkage levels revealed on an interim basis from the newly implemented cycle count approach were projected to year end, yet shrinkage rates were increasing during the period.

Evidence was also provided in the case that suggested the presence of significant inventory marketability problems. Sales were flat, inventory was up, and competition in the industry was intense. Furthermore, inventory was significantly overstated due to improper purchases in the west coast stores (a new geographic market area) and heavy industry discounting. These conditions resulted in the need to write-down a material amount of inventory to the lower-of-cost-or market.

The last error was a generic expense payables cutoff error. Selling expense was flat, yet the background information noted increased advertising efforts. Selling expense and related payables were significantly understated due to a cut-off/accrual error resulting from weak controls at the stores.

Also embedded in the case was an industry-specific "blind alley" or area of low exposure. Accounts receivable increased significantly, suggesting potential bad debt problems. However, this change is generally not a problem in retailing where, as the case here, receivables are from major credit cards and/or sold for a processing fee.

Along with the background and financial data, subjects were provided with a preliminary time budget based on the average hours encountered in each of twenty-two account and planning categories for a typical retailing client of the size represented in the case (annual net revenue of $2.3 billion; total assets $1.1 billion). Without a preliminary time budget, analysis of the data is likely to suffer from considerable variance in responses due to the lack of a comparable starting point (Cohen and Kida, 1989). The preliminary time budget was developed by the engagement partner who was specifically directed to prepare a time budget for a retailing client of the size indicated in the case where there were no material errors present. As for a typical engagement, the field work (hours) in testing the accounts was divided among staff and seniors with the greater proportion assigned to the former. The initial budget is considered to be a neutral starting point in that it did not allow additional time for the identified audit errors. Further, the neutrality of the budget was reviewed through discussions with retailing partners and extensive pilot testing. This phase also revealed that additional hours were perceived as necessary to properly test the misstated accounts and for additional supervision
and review.

Independent Variables

The independent variables were industry and general experience. General audit experience (years of audit experience and staff level--partner, manager, and supervisor/senior) was captured to control for this variable in testing the research hypotheses. Since there is no variable that is recognized as the optimal measure of industry specialization, subjects were asked to report four measures of industry experience: number of retailing audit engagements; number of retailing clients; hours of staff training directed specifically toward the retailing industry; and years of non-public accounting experience directly in retailing. Retail engagements are the number of audits involved in, while retail clients relate to separate

companies. Thus, a repeat audit would represent two engagements but only one client. The alternative measures examined were suggested in discussions with audit partners while developing and pretesting the case[6].

The Tasks

As noted previously, subjects were asked to perform three planning tasks: identification of likely errors; assessment of the likelihood of material error by account; and determination of planned audit hours by staff level.
Each of these tasks will be described in detail.

Identification of Likely Errors Present (H1-H2)

To test the impactof industry experience on hypothesis generation, subjects were asked to indicate the type and cause of material error(s) that they believed may be present given the information in the case and their own analytical procedures. Subjects were encouraged to identify as many errors as they believed existed. Responses were coded by two independent persons, each with five years of audit experience, utilizing the following categories:

1 Cycle count (increasing shrinkage)
2 Markdowns (buyers initiate markdowns at store)
3 Net-realizable value (obsolescence)
4 Expense payable cutoff
5 Accounts payable
6 Bad debt provision
7 Other errors
8 General error
9 Incorrect error

The first four categories represent the errors present in the case. An increase in accounts receivable and the introduction of an automated accounts payable system led several auditors in an initial random sample of responses (n=10) to identify these two error types as likely to be present in the case. Thus, coding categories were also established for these error types. "Other" errors are those not related to the categories above such as a sales cut-off problem. A "general" error refers to instances where a subject noted a problem but was not specific enough to

identify any particular type of error, e.g. "sales look overstated". Finally, an incorrect error is when the hypothesis noted was inconsistent with the facts in the case such as "an inventory understatement due to failure to include all inventory owned" when, in fact, the inventory balance increased over 50%. The level of interrater agreement was 81.2%. Coding discrepancies were then reconciled by one of the authors.

Assessment of the Likelihood of Material Error & Planned Extent

The likelihood of material error was elicited by account on a 10 point likert scale (5 points above and below the mid-point) with verbal end anchors similar to that used in practice indicating low, moderate, and high risk. Subjects were also told that for the purpose of assessing the likelihood of material error, materiality was assumed to be 5% of average pretax income or $4 million; all of the seeded errors exceeded this threshold.

The time budget identified the audit testing strategy in each area: control testing or substantive testing. Subjects were told that the orientation was based on a preliminary review of controls. This information provides control over potential confounding factors such as differences in auditors' control risk assessments and control testing propensities as well as the relative cost of control testing versus substantive testing.

Performance Benchmarks and Interactive Effects

The presence of the four material errors in the case provided a benchmark against which to evaluate each subject's ability to generate hypotheses, judge the likelihood of error, and to plan the extent of testing. For the accounts containing a material misstatement(s) planned hours are expected to be higher and focused more towards experienced staff than contained in the budget.

Given the exploratory nature of the research, interactive effects between general and industry experience are not hypothesized. However, interactive effects are plausible. For instance, an auditor with both high levels of general and industry experience may be particularly adept in making the planning judgments examined here. Potential interactive effects are, thus, investigated in the analyses.

ANALYSIS AND RESULTS

Regression was the primary method of analysis used to test the hypotheses. The general regression model employed was as follows:

$$\text{Audit judgment} = a + b_1 \text{Exper} + b_2 \text{Indus} + b_3 (\text{Exper} \times \text{Indus}) + e$$

Where: Exper= general audit experience (years)
Indus= industry experience (number of retailing engagements)

Years of audit experience was employed as the measure of general audit experience, since staff level and years of experience were found to be highly correlated ($r=.842$; $p<.001$). Further, the results focus on the number of retailing engagements as the measure of "industry experience", since the number of retailing engagements and the alternative variables were highly correlated (retailing clients-- $r=.750$ and hours of retailing training-- $r=.564$). Also the findings were the same using these alternative variables.

Hypotheses Generation (H1 & H2)

H1 & H2A posit that auditors with greater industry experience will generate a larger number of plausible hypotheses and be more likely to identify the industry-specific errors present. However, no differences in hypothesis generation are expected in identifying the presence of a generic error (H2B). Plausible hypotheses were considered to be all categories except a general error or an incorrect error.

The results for the hypotheses generation task are presented in Table 2. As expected, industry experience enhanced hypotheses generation as noted by the significant positive regression coefficient for 'plausible hypotheses' ($p=.003$). Subjects with industry experience were also able to better identify the industry-specific errors present ($p=.005$). These results provide support for H1 and 2A.

Marginally significant interactions were also found between general and industry experience and the number of plausible ($p=.07$) and industry errors ($p=.03$). To explore the

nature of these interactions, the sample was partitioned into low and high levels of industry and general experience, operationalized as the top and bottom 20% of each distribution. The exception is low industry experience, since a large proportion (n=27) of subjects had no retailing experience at all. The mean number of hypotheses generated for each group is reported at the bottom of Table 2, indicating that greater industry experience significantly ($p < .10$) enhanced hypothesis generation while general experience had no significant effect.[7]

Also as indicated in Table 2, industry experience did not enhance the auditors' ability to generate the generic error, an improper expense payable cutoff. This findings supports H2B.

To further explore the effect of industry experience on hypothesis generation, logit analyses were conducted for each specific error type. The model employed was the same as that used in the regression analyses, except that the dependent variable was whether or not the specific error was generated. The findings indicated that industry experience significantly ($p < .05$) enhanced the generation of two of the three retailing errors embedded in the case (markdowns and net-realizable value) but not the general expense payable error, as expected. There was also a marginally significant positive association ($p= .086$) between general audit experience and the identification of the net-realizable value error. Additionally, industry experience also led to a significantly higher likelihood of generating the plausible hypothesis that there was a misstatements in accounts payable ($p=.061$).

Likelihood (Risk) of Material Error by Account (H3)

In H3 it was hypothesized that auditors with industry experience would be better able to assess the risks associated with accounts impacted by industry specific errors. As indicated in Table 3, the accounts affected by the retailing errors were inventory and cost of sales, while the generic error impacted expense payable and selling expenses. Additionally, the findings for the "blind alley" in the case (accounts receivable) are shown.

Contrary to expectations, the results in Table 3 do not indicate a significant effect for industry experience in risk assessments for the accounts misstated by the retailing errors. Industry experience did, however, marginally impact ($p \leq .10$) risk assessments for the two accounts relating to the non-retailing error but the direction of the relationship for selling

expense was opposite (negative) of expectations. Finally, there was no significant industry effect for accounts receivable. Thus, the findings do not support H3.

There was also lack of support for the impact of general audit experience on risk assessments. In addition, significant interactions between general audit and industry experience were not found.

Efficiency and Effectiveness in Program Planning (H4 & H5)

H4 and H5 posit that industry experience enhances audit effectiveness and efficiency in terms of planned extent. To test these hypotheses, general audit and industry experience were regressed against revised audit hours (i.e., planned versus budget). Audited hours are expected to be increased (exceed budget) when errors are present (H4); a higher percentage of hours is also hypothesized than budgeted for more experienced staff in such cases (H5). The results are presented in Tables 4 and 5.[8]

As indicated in Table 4, industry experience did not have a significant effect on revised audit hours by account. Further, general audit experience led to significantly greater audit hours than budgeted across many accounts (inventory observation, inventory aggregate, selling expense, and accounts receivable) as well as increased administrative and total engagement hours. This finding suggests greater conservatism in planning audit efforts by experienced auditors (Taylor, 1994). Finally, significant interactions were not present between general audit and industry experience. Overall, these findings do not support H4.

Hypothesis 5 is partially supported. As shown in Table 5, except for inventory observation, industry experience significantly affected staff assignments, as expected. Ceteris parabis, industry experience was anticipated to be associated with a decrease in the percentage of hours planned for less experienced (staff) personnel assigned to industry-specific error accounts, i.e., negative beta coefficients in the regression results. Proportionally fewer hours were also found for less experienced personnel for the engagement overall. Unexpectedly, though, lower extent was also planned for staff assigned to the accounts affected by the generic error in the case. No significant findings were present for accounts receivable. Potential reasons for these findings are discussed in the following section. Finally, there were no

significant effects found regarding staff assignment relating to general experience or the interaction between general and industry experience.

DISCUSSION

The results indicate that industry experience enhanced hypothesis generation in the planning phase, which would be expected to positively impact the effectiveness and efficiency of subsequent audit testing (information search) and ultimately lead to higher decision performance. Nonetheless, industry experience did not impact risk assessments as to the likelihood of material error in account areas or planned hours. The assignment of hours to various staff levels was, however, affected by industry experience for accounts but at times not as expected. That is, a greater proportion of time for experienced staff was anticipated for industry-specific account areas of high likelihood of error and vice versa for accounts of low exposure. Instead, although the former expectation was confirmed by the findings, more experienced staff were also assigned to accounts affected by a generic error. Perhaps these responses represent areas of the audit that may contain errors for many retailing clients as learned through industry experience and, thus, represent broad planning heuristics. Future research is needed to test the validity of this potential explanation, and, if present, to assess whether such heuristics lead to greater efficiency or effectiveness across a spectrum of clients.

The overall findings raise the question of why auditors with greater industry experience were apparently better able to generate plausible errors and identify the errors present in the case but, contrary to expectations, did not demonstrate improved accuracy in assessing the likelihood of material error for individual accounts. These results may be because the error frequency knowledge of experienced auditors is not organized by account but by error type (Tubbs 1992; Frederick 1991); thus, account error frequencies may be less retrievable (Nelson 1994). Additionally, errors affect at least two accounts and assigning likelihood assessments to individual accounts may be a difficult task since auditors store error information by objectives and not by accounts (Bonner et. al. 1996a; Nelson et. al. 1995). Thus, the task demands of estimating misstatement risks at the account level may not match the nature of the underlying knowledge structure.

Ashton (1991) also reports that greater general audit and industry experience did not lead to increased accuracy in assessing population account error frequency rates but did result in greater accuracy in identifying frequently occurring error types. She attributes these results to the fact that the occurrence of material errors is relatively rare, and auditors receive incomplete feedback of error frequencies across accounts as a result of limited exposure to a variety of clients. Thus, auditors may focus on error types rather than account error frequencies. While Ashton examined auditor accuracy in assessing population error rates, this study investigates auditors' diagnostic skills for a particular case. Industry experience was found to enhance an auditor's knowledge of error types, which should be functional in identifying areas of exposure and planning audit tests. The types of errors present in the case examined (markdowns, inventory shrinkage, marketability, and expense payable cutoff) are relatively frequent for retailing companies. Thus, the findings here for an individual case corroborate those of Ashton. Future research is needed to evaluate these and other potential explanations for the unexpected findings here regarding the impact of industry experience on risk assessments.

Given greater apparent hypothesis generation skills, it is also surprising that there was no significant impact of industry experience on planned audit extent (hours). Presumably recognition of likely errors would lead to greater extent in areas of exposure and reduced extent in areas of low risk. These results mirror those of an archival study by Mock and Wright (1993) who report a weak association between the level of and changes in risk and accompanying program planning decisions. Alternatively, despite careful pilot testing, perhaps the time budget was not, in fact, considered “neutral” with respect to errors and was viewed as providing sufficient hours in the risk areas such as inventory. Finally, audit hours were elicited before the hypothesis generation task, which may have made subjects less sensitive in planning hours to potential errors present. Using student subjects, Bonner et. al. (1996b) report that error frequency knowledge affected the allocation of hours only when error frequency estimation preceded this audit judgment, suggesting individuals need to be prompted to apply frequency knowledge. However, as discussed earlier, the order of tasks in the experiment was done to avoid potentially biasing planning judgments by explicit hypothesis generation, which is not normally done in practice. Overall, there are few studies on the broad issue of whether auditors

sufficiently adapt program plans in response to recognized risks, a promising area for future research.

This study examined the relationship between risk assessments and extent but did not address decisions as to nature and timing of tests. Industry experience could potentially be very beneficial for these planning decisions, an important avenue for future research.

General audit experience was found to lead to increased audit extent but both for error and non-error ("blind-alley") accounts. Perhaps this result reflects the general sensitivity of experienced auditors to the greater risks present for the rapidly growing client in the case, facing a competitive, changing market. Thus, extent was increased across the board for many account areas, suggesting greater conservatism. Taylor (1994) also reports increased conservatism for auditors lacking industry experience. Such conservatism, while potentially enhancing audit effectiveness, may, however, substantially reduce efficiency.

The limited research regarding the affects of industry experience suggests this is a significant area for future study. For instance, we have no evidence on how industry knowledge impacts information search or other audit tasks such as evidence evaluation. Hopefully this study will spur further efforts in this important area of research.

FOOTNOTES

[1] There were 62 subjects obtained from one firm and 10 from another. The results did not vary between firms. Therefore, aggregate findings are reported.

[2] Three demographic responses were omitted from the data reported in Table 1 due to apparent confusion or misunderstanding as to the nature of the question. One individual indicated experience with 12 retailing clients, even though he had not been on any retail engagements. Another person indicated more retail clients than engagements, and finally one other indicated an extreme number of hours of training.

[3] The financial data included key statistics and ratios as well as a Balance Sheet and Income Statement. The financial statistics included ratios that are widely used in the retailing industry such as sales per square foot, markups %, markup cancellations %, markdowns %, markdown cancellations %, and inventory shrinkage % (%'s of sales).

[4] Ten auditors participated in the pretest, distributed by staff level and industry background as follows:

	Retailing	Non-retailing
Partner	2	0
Manager	3	1
Senior	2	1
Staff	0	1
Total	7	3

Discussions with partners at the participating firm indicated that auditors with five or more retailing engagements are considered to have very extensive background in the industry. The case discussion was modified and examples were added to clarify the response scales based on pretest feedback.

[5] The audit adjusting entry for each error was as follows:

1.markdown omissions:

Cost of sales	$5 million	
Inventory		$5 million

2. inventory shrinkage:

Cost of sales	$6 million	
Inventory		$6 million

3. lower-of-cost-or market:

Cost of sales	$5 million	
Inventory		$5 million

4. improper expense payables cutoff:

Selling expenses	$7 million	
Expense payables		$7 million

Materiality was established at $4 million, which was about 5% of average pretax profit.

[6] Data were also gathered on an additional measure of retailing experience: years of non-public accounting experience directly in the retail industry. However, 81% of the participants did not have such experience (mean .5 years), and there was no significant difference along this dimension between the retailing and non-retailing groups. Thus, this factor was not used in subsequent analyses.

[7] To test the robustness of these findings, the sample was also partitioned in two other ways: lowest and highest third; and lowest and highest quarter. The findings of comparison t-tests were the same as those reported in Table 2 with only one exception: when partitioned into thirds, the high industry experience group, although identifying more industry-specific errors, did not generate a significantly different number of hypotheses (p=.11).

[8] Regression analyses were also conducted using planned hours and percentage of staff time directly. The findings were the same as those reported in Tables 4 and 5.

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TABLE 1

DEMOGRAPHIC DATA ON SAMPLE

	RETAIL (n=34)		NON-RETAIL (n=38)		
	(mean)	range	(mean)	range	t
Years of general audit experience (EX)	8.79 (4.50)	3-20	8.07 (7.31)	2-30	.50
Number of retail engagements (RE)	7.35 (4.53)	3-18	.47 (.80)	0-2	9.22**
Number of retail clients (RC	3.52 (2.12)	1-10	.32 (.58)	0-2	8.79**
Hours of retail training (RT)	43.06 (47.70)	10-20	5.84 (13.12)	0-48	4.56**

**p<.001

Position in firm (n):	TOTAL	RETAIL	NON-RETAIL
partner	15	6	9
manager	31	19	12
supervisor/senior	26	9	17
Total	72	34	38

$\chi^2 = 4.434$ (df = 2; ns)

NOTES: (1) Retailing subjects are those with significant industry experience (3 or more engagements).
(2) Standard deviations are indicated in parenthesis.

TABLE 2

(n=72)

		GENERAL AUDIT EXPERIENCE			INDUSTRY AUDIT EXPERIENCE			INTERACTION			
Dep. Var.	Constant	Beta Coef.	t	Sign.	Beta Coef.	t	Sign.	Beta Coef.	t	Sign.	R^2
Plausible Hypoth. (H1)	4.21	-0.001	.02	ns	0.41	2.85	.003	-.0248	1.86	.07	.15
Industry Error (H2A)	1.40	0.03	1.35	ns	0.18	2.69	.005	-.0137	2.24	.03	.10
Generic Error (H2B)	0.28	0.002	.14	ns	0.02	0.46	ns	.07	.09	ns	.06

DESCRIPTIVE DATA ON NUMBER OF HYPOTHESES GENERATED

	Low general experience (n=16)	High general experience (n=15)	Low industry experience (n=27)	High industry experience (n=15)
Plausible Hypothesis (H1)				
mean	4.69	4.20	4.11**	6.13**
S.D.	2.62	1.70	1.92	2.85
Industry Errors (H2A)				
mean	1.56	1.67	1.67*	2.27*
S.D.	.81	.98	1.04	1.44

NOTES: (1)Low general experience is less than or equal to 3 years. High experience is equal to or more than 12 years. Low industry experience is equal to 0 engagements. High is equal to or greater than 7 engagements.
(2)These results represent one tail tests consistent with hypothesized directional expectations.
(3)p values shown where $\leq$.10.

T-tests; comparisons between 1) low and high general experience and 2) low and high industry experience.
**p<.05 (one-tailed test)
* p<.10 (one-tailed test)

TABLE 3

AUDITOR RISK ASSESSMENTS (H3)
(n=72)

		GENERAL AUDIT EXPERIENCE			INDUSTRY AUDIT EXPERIENCE			
Error Account Area:	Constant	Beta Coef.	t	Sign.	Beta Coef.	t	Sign.	R^2
RETAILING ERRORS								
Inventory	8.13	-.036	.09	ns	-.020	.20	ns	.02
Cost of sales	6.72	.026	.59	ns	.091	.77	ns	.01
GENERIC ERROR								
Expense payable	5.96	-.008	.15	ns	.196	1.42	.08	.04
Selling expense	5.99	-.076	1.58	ns	-.246	1.92	.06	.07
BLIND ALLEY								
Acc. Receivable	4.71	.028	.63	ns	.053	.41	ns	.008

NOTES: (1) Dependent variable is likelihood of a material error: 1-10 (Low to High)
(2) These results represent one-tail tests consistent with hypothesized directional expectations.

TABLE 4

PLANNED LEVEL OF TESTING (H4)
(n=72)

		GENERAL AUDIT EXPERIENCE			INDUSTRY AUDIT EXPERIENCE			
Error Account Area:	Constant	Beta Coef.	t	Sign.	Beta Coef.	t	Sign.	R^2
RETAILING ERRORS								
Inventory observ.	96.89	1.50	1.38	.09	1.21	0.42	ns	.03
Inventory sum.	215.08	1.77	1.19	.23	-1.04	0.26	ns	.11
Inventory aggr.	312.00	3.28	1.54	.07	0.18	0.03	ns	.07
Cost of sales	69.31	1.31	1.73	.04	0.13	0.07	ns	.06
GENERIC ERROR								
Expense payable	64.87	0.14	0.31	.76	1.12	0.92	ns	.10
Selling expense	45.91	1.49	3.08	.001	0.49	0.38	ns	.14
BLIND ALLEY								
Accounts receivable	44.46	1.03	2.02	.05	-0.79	0.58	ns	.10
Total admin. Hours	721.21	4.49	1.44	.07	-5.86	0.71	ns	.07
Total Engagements	1699.53	15.01	2.14	.04	-5.58	0.30	ns	.12

NOTES: (1) Dependent variable is Planned Hours - Budgeted Hours; the expectation is that planned hours would be expanded (exceed the budget) where material misstatements are likely to be present.
(2) These results represent one tail tests consistent with hypothesized directional expectations.

TABLE 5

ASSIGNMENT OF HOURS TO AUDIT STAFF (H5)
(n=72)

		GENERAL AUDIT EXPERIENCE			INDUSTRY AUDIT EXPERIENCE			
Error Account Area:	Constant	Beta Coef.	t	Sign.	Beta Coef.	t	Sign.	R^2
RETAILING ERRORS								
Inventory observ.	.78	0.0005	.36	.72	0.002	.54	.59	.10
Inventory sum.	.78	-.003	.93	.36	-0.02	2.38	.02	.08
Inventory aggr.	.78	-0.002	.82	.41	-0.013	1.99	.05	.07
Cost of sales	.79	-0.004	1.17	.24	-0.02	2.01	.05	.06
GENERIC ERROR								
Expense payable	.82	-0.001	.45	.66	-0.02	3.54	.001	ns
Selling expense	.79	0.001	.55	.58	-0.01	2.37	.02	ns
BLIND ALLEY								
Accounts receivable	.79	-0.002	.80	.43	-0.0004	.07	.95	.03
Total Engagement	.46	-0.0008	.07	.95	-0.005	1.48	.14	.06

NOTES: (1) Dependent variable is Percentage of Hours assigned to less experienced staff less Budgeted Percentage; i.e., change in proportion of hours assigned to staff. The expectation is that this percentage will decline for accounts likely to be misstated.
(2) These results represent one tail tests consistent with hypothesized directional expectations.
(3) p values shown where $\leq .10$.